

Hypoxia-inducible factor prolyl 4-hydroxylase-2 inhibition protects from retinopathy of prematurity

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Background: Retinopathy of prematurity (ROP) is the leading cause of childhood blindness. High O₂ levels (hyperoxia) during early neonatal life is the key risk factor for ROP. Current therapies primarily target the late vasoproliferative phase of the disease (Phase II), whereas no interventions are available for the early phase (Phase I). Accumulating evidence suggests that hypoxia-inducible factor (HIF) signaling and the O₂-regulated HIF prolyl 4-hydroxylases (HIF-P4H) are centrally involved in the pathophysiology of ROP. Thus, timed downregulation of HIF-P4Hs in hyperoxia, resulting in HIF activation, is a promising treatment option in ROP.

Methods: We utilized the oxygen-induced retinopathy (OIR) mouse model to evaluate whether HIF activation through HIF-P4H-2 inhibition protects against preclinical ROP. Mice with partial deficiency of HIF-P4H-2 (*Hif-p4h-2^{gt/gt}*) and their wild-type littermates were subjected to OIR model to assess susceptibility to retinovascular pathology. Retinal HIF1 α expression levels and HIF target gene expression were measured and compared between *Hif-p4h-2^{gt/gt}* mice and wild-type littermates in different timepoints. Serum insulin-like growth factor-1 (IGF-1) and erythropoietin (EPO) levels were measured in normoxic and hyperoxic conditions.

Results: *Hif-p4h-2^{gt/gt}* mice were protected against microvascular obliteration in Phase I, followed by less neovascularization and preserved arterial morphology in Phase II of the OIR model. *Hif-p4h-2^{gt/gt}* mice showed 3-fold higher levels of retinal HIF1 α in hyperoxia in Phase I accompanied with lower expression of *Hif-p4h-3* in Phase II. In addition, *Hif-p4h-2^{gt/gt}* mice showed 2–3 fold higher levels of serum EPO throughout neonatal period both in normoxic and hyperoxic conditions. Although *Hif-p4h-2^{gt/gt}* mice were leaner, the weight difference persisted similar throughout the experiment.

Conclusions: Partial deficiency of HIF-P4H-2 confers protection against hyperoxia-induced microvascular obliteration and mitigates subsequent pathological neovascularization in the OIR model. Early activation of the HIF pathway appears to enhance retinal resilience to hyperoxic injury, including the selective transcriptional response in retina and systemic increase in EPO levels without compromising postnatal weight gain. These findings support HIF-P4H-2 inhibition as a promising Phase I-targeted therapeutic strategy for ROP.